Algebraic Data Types with Sealed Records

Exhaustive Payment describe

Season 6 · Functional · Intermediate

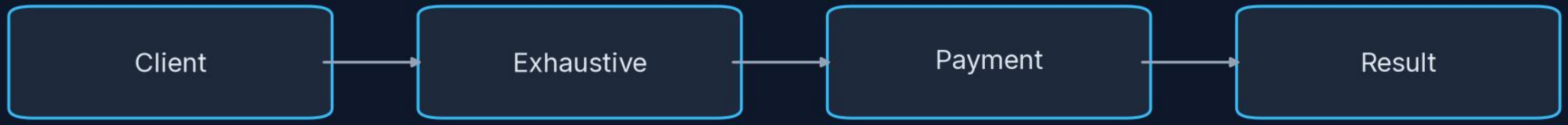
Today's Hook

- Problem: Exhaustive Payment describe
- Naive pain cue: Failure path is asserted in ProblemTest (injected fault / invalid input)
- We will refactor to ADTs Sealed with a runnable Demo.
- Failure path is locked by ProblemTest.

The Problem

- Scenario: Exhaustive Payment describe
- Constraints: solve «Exhaustive Payment describe» with ADTs Sealed — keep the demo runnable and the failure path observable.
- Naive pain: Naive version fails to cleanly solve: Exhaustive Payment describe
- Success looks like: card:10

Structure



ADTs Sealed: Exhaustive Payment describe

Real-World Scenario

- Running example: Exhaustive Payment describe
- Key types: Payment, Card, Cash, Wire
- Happy-path stdout cue: card:10
- Failure-path stdout cue: Failure path is asserted in ProblemTest (injected fault / invalid input)

Naive Code

```java
package com.buildwithwaqar.patterns.functional.adt.problem;

import java.util.*;

public final class NaivePay {
  public static String describe(String type, int cents){
    if("card".equals(type)) return "card:"+cents;
    if("cash".equals(type)) return "cash:"+cents;
    return "other";
  }
}
```

Why Naive Hurts

- Naive version fails to cleanly solve: Exhaustive Payment describe
- Without ADTs Sealed, call sites absorb the wrong details.
- Failure mode to show on camera: Failure path is asserted in ProblemTest (injected fault / invalid input)
- New requirements force shotgun edits across the naive code.

Solution

```java
package com.buildwithwaqar.patterns.functional.adt.problem;

import java.util.*;
import java.util.function.*;

sealed interface Payment permits Card, Cash, Wire {}
record Card(String last4, int cents) implements Payment {}
record Cash(int cents) implements Payment {}
record Wire(String iban, int cents) implements Payment {}
final class Payments {
  static String describe(Payment p){
    return switch (p) {
      case Card(String last4, int cents) -> "card ****"+last4+" "+cents;
      case Cash(int cents) -> "cash "+cents;
      case Wire(String iban, int cents) -> "wire "+iban+" "+cents;
    };
  }
}
```

Runnable main

```
package com.buildwithwaqar.patterns.functional.adt.problem;
import java.util.*; import java.util.function.*;
public final class ProblemDemo {
  public static void main(String[] args) {
    System.out.println("=== ADTs: exhaustive Payment describe ===");
    System.out.println("--- naive ---"); System.out.println(NaivePay.describe("card", 10));
System.out.println("--- ADT ---");
System.out.println(Payments.describe(new Card("1234", 1999)));
System.out.println(Payments.describe(new Cash(500)));
System.out.println(Payments.describe(new Wire("DE89", 10000)));
  }
}
```

Walkthrough Output

```
=== ADTs: exhaustive Payment describe ===
--- naive ---
card:10
--- ADT ---
card ****1234 1999
cash 500
wire DE89 10000
```

Step-by-Step Trace

- Match stdout to stages — look for: card:10
- Then the injected failure: Failure path is asserted in ProblemTest (injected fault / invalid input)
- Re-run: java -cp … 53-adts-sealed ProblemDemo (see episode folder).
- Pause the video and predict the next printed line.

Modern Java 25 / Zazr

```java
package com.buildwithwaqar.patterns.functional.adt.problem;

import java.util.*;
import java.util.function.*;

sealed interface Payment permits Card, Cash, Wire {}
record Card(String last4, int cents) implements Payment {}
record Cash(int cents) implements Payment {}
record Wire(String iban, int cents) implements Payment {}
final class Payments {
  static String describe(Payment p){
    return switch (p) {
      case Card(String last4, int cents) -> "card ****"+last4+" "+cents;
      case Cash(int cents) -> "cash "+cents;
      case Wire(String iban, int cents) -> "wire "+iban+" "+cents;
    };
  }
}
```

Trade-offs

- Use ADTs Sealed when «Exhaustive Payment describe» repeats and a simpler feature does not erase it.
- Refuse extra types when a sealed switch or a function value already suffices.
- Teaching demos omit production metrics, retries, and multi-region clocks.
- Interview soundbite: state the force behind ADTs Sealed in product language.

Related Patterns

- Name the pattern usually confused with ADTs Sealed and separate the forces.
- Sketch the roles for «Exhaustive Payment describe» from memory in sixty seconds.
- Point to the next episode in this season on the playlist map.
- Link this episode's series/ folder in the YouTube description.

Recap

- Problem: Exhaustive Payment describe
- Naive pain → ADTs Sealed boundary → Demo stdout.
- Remember the failure cue: Failure path is asserted in ProblemTest (injected fault / invalid input)
- Break a ProblemTest assertion, watch it fail, restore it.